Circles

Class 2

1 Finding the centre of a circle

Jamuna has drawn a circle by tracing around a bangle. She wants to find its centre. Her friend Amina says, "Fold the paper twice." Solve P1 to see why this works.

Solve first.

P1. Trace around a bangle on an A4 sheet. Draw any chord AB that is not a diameter. Hold the sheet against a window and fold it so that A falls exactly on B. Open the sheet.

(a) The crease is the perpendicular bisector of AB. Which special point of the circle must it pass through? Use a fact from Class 1.

(b) Draw a second chord CD, not parallel to AB, and fold again so that C falls on D. Where do the two creases meet?

(c) Why is one crease not enough to find the centre?

(d) What goes wrong if CD is parallel to AB?

Things to Remember

To find the centre of a circle, draw two chords that are not parallel. Their perpendicular bisectors meet at the centre.

2 Circles through two points

Solve first.

P2. Points A and B are 8 cm apart, and M is the midpoint of AB. Draw a rough diagram for each part.

(a) Find the centre and the radius of the smallest circle that passes through A and B.

(b) A circle of radius 5 cm passes through A and B, and its centre is O. Explain why O lies on the perpendicular bisector of AB. Find OM.

(c) How many circles of radius 5 cm pass through A and B?

(d) Can a circle of radius 3 cm pass through A and B? Give a reason.

P3. Use P2 to answer these questions.

(a) Where do the centres of all the circles through A and B lie?

(b) As the centre moves away from M, does the radius increase or decrease?

(c) Is there a largest circle through A and B?

Things to Remember

Infinitely many circles pass through two points A and B. Their centres lie on the perpendicular bisector of AB. The smallest has AB as a diameter.

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3 Circles through three points

Solve first.

P4. A, B and C lie on a straight line, with $AB = 4$ cm and $BC = 6$ cm.

(a) Draw the perpendicular bisectors of AB and BC.

(b) Do they meet? Give a reason.

(c) Can a circle pass through A, B and C?

P5. Draw any triangle ABC on an A4 sheet.

(a) Construct the perpendicular bisectors of AB and BC. Mark their meeting point O.

(b) Measure OA, OB and OC.

(c) Explain, without measuring, why $OA = OB = OC$.

(d) Does the perpendicular bisector of CA also pass through O? Why?

Theorem 5 (NCERT Theorem 1). *There is a unique circle passing through three non-collinear points.*

FIG

Why it works.

Given: three points A, B and C that are not on one line.
To show: exactly one circle passes through A, B and C.

1. The centre of any circle through A and B lies on the perpendicular bisector of AB (P3). The centre of any circle through B and C lies on the perpendicular bisector of BC.
2. AB and BC are not on one line, so their perpendicular bisectors are not parallel. They meet in exactly one point O.
3. $OA = OB = OC$, so the circle with centre O and radius OA passes through all three points.
4. Any circle through the three points must have its centre at this point O and radius OA. So there is only one such circle.

This circle is the **circumcircle** of $\triangle ABC$. Its centre O is the **circumcentre**, and its radius is the **circumradius**.

Solve first.

P6. Construct each triangle and its circumcircle. Say whether the circumcentre lies inside the triangle, on a side, or outside the triangle.

(a) $AB = 5$ cm, $\angle A = 70°$, $\angle B = 60°$.

(b) $PQ = 6$ cm, $\angle P = 90°$, $PR = 8$ cm.

(c) $XY = 5$ cm, $\angle X = 100°$, $XZ = 4$ cm.

Things to Remember

The circumcentre of a triangle lies inside an acute-angled triangle, at the midpoint of the hypotenuse of a right-angled triangle, and outside an obtuse-angled triangle.

Example 1. A potter finds a broken piece of a round clay plate. Explain how to find the centre and the radius of the full plate.

FIG

Given: a piece of the rim of a circle.
To find: the centre and the radius of the circle.

1. Place the piece on paper and trace its curved edge. Mark three points A, B and C on the traced arc.
2. Draw the chords AB and BC, and construct their perpendicular bisectors. They meet at O.
3. O is the centre: the perpendicular bisector of every chord passes through the centre, and only one circle passes through A, B and C (Theorem 5).
4. Measure OA. This is the radius.

Answer: the centre is the point where the two perpendicular bisectors meet, and the radius is OA.

Exercise 2A

1. Construct $\triangle ABC$ with $AB = 6$ cm, $BC = 7$ cm and $CA = 7$ cm. Construct its circumcircle, and let the circumcentre be O. Measure OA, OB and OC.
2. Points A and B are 10 cm apart. What is the least possible radius of a circle through A and B? Is there a greatest possible radius? Explain.
3. Points P and Q are 6 cm apart. A circle of radius 5 cm passes through P and Q.
   (a) Find the distance of its centre from the line PQ.
   (b) How many such circles are there?
   (c) Explain why no circle of radius 2 cm passes through P and Q.
4. A, B and C lie on a straight line. Is there a point P with $PA = PB = PC$? Explain using the perpendicular bisectors of AB and BC. Can a straight line meet a circle in three points?
5. Three villages lie at the corners of a triangle. The panchayat wants to build a water tank at the same distance from all three. Explain how to find the place on a map. What happens if the three villages lie on one straight road?
6. Kiran wants to find the centre of a circular plate drawn on paper. He draws two parallel chords and folds the sheet so that the ends of each chord meet. Find his mistake, and correct the method.

7. The circumcircle of $\triangle ABC$ is drawn. Can another triangle, congruent to $\triangle ABC$, have the same circumcircle? Explain.

8. The sides of a right-angled triangle are 5 cm, 12 cm and 13 cm. Where is its circumcentre? Find the circumradius.

9. The circumcentre of $\triangle ABC$ lies on the side BC. What kind of triangle is $\triangle ABC$? Draw a rough figure to support your answer.

4 Chords and their distance from the centre

Recall from Class 1: the distance of a chord from the centre is the length of the perpendicular from the centre to the chord.

Solve first.

P7. A circle has centre O and radius 5 cm. Chords AB and CD are each 8 cm long. $OM \perp AB$ and $ON \perp CD$.

(a) Find OM and ON.

(b) What do you notice?

(c) Would the answer to (b) change for two other equal chords of this circle? Why?

P8. In the same circle, chords PQ and RS are each 4 cm from O. Find PQ and RS.

P9. In the same circle, copy and complete the table.

Distance from O (cm)	0	3	4	5
Length of chord (cm)				

(a) As the chord moves away from the centre, what happens to its length?

(b) Which chord is the longest? Where does it lie?

(c) Of two unequal chords, which one is nearer the centre?

Theorem 6 (NCERT Theorem 6). *Chords of a circle having the same length are all at the same distance from the centre of the circle.*

FIG

Why it works.

Given: chords AB and CD of a circle with centre O, $AB = CD$, $OM \perp AB$ and $ON \perp CD$.
To show: $OM = ON$.

1. M and N are the midpoints of AB and CD (Theorem 4). Since $AB = CD$, $AM = CN$.
2. In $\triangle OMA$ and $\triangle ONC$: $\angle OMA = \angle ONC = 90°$, $OA = OC$ (radii) and $AM = CN$.
3. So $\triangle OMA \cong \triangle ONC$ (RHS), and $OM = ON$.

Theorem 7 (NCERT Theorem 7). *Chords of a circle that are equidistant from the centre have equal length.*

Why it works.

Given: chords AB and CD of a circle with centre O, $OM \perp AB$, $ON \perp CD$ and $OM = ON$ (same figure).
To show: $AB = CD$.

1. In $\triangle OMA$ and $\triangle ONC$: $\angle OMA = \angle ONC = 90°$, $OA = OC$ (radii) and $OM = ON$ (given).
2. So $\triangle OMA \cong \triangle ONC$ (RHS), and $AM = CN$.
3. $AB = 2AM$ and $CD = 2CN$ (Theorem 4), so $AB = CD$.

Theorem 8 (NCERT Theorem 8). *Let AB and CD be two chords of a circle with centre O, with $AB > CD$. Then the distance of AB from O is less than the distance of CD from O.*

Why it works.

Given: $AB > CD$, $OM \perp AB$ and $ON \perp CD$ (same figure, with AB longer).
To show: $OM < ON$.

1. By Pythagoras theorem, $OM^2 + AM^2 = OA^2$ and $ON^2 + CN^2 = OC^2$. Since $OA = OC$, $OM^2 + AM^2 = ON^2 + CN^2$.
2. $AM = \dfrac{AB}{2}$ and $CN = \dfrac{CD}{2}$. Since $AB > CD$, $AM > CN$, so $AM^2 > CN^2$.
3. The two sums are equal, so $OM^2 < ON^2$, and $OM < ON$.

Things to Remember

For a chord at distance d from the centre of a circle of radius r, the length of the chord is $2\sqrt{r^2 - d^2}$. Equal chords are at equal distances from the centre, and the longer of two chords is nearer the centre. The diameter is the longest chord.

Example 2. A circle has centre O and radius 17 cm. Chord AB is 30 cm long and chord CD is 16 cm long. Which chord is nearer the centre? Find the distance of each chord from O.

FIG

Given: $r = 17$ cm, $AB = 30$ cm, $CD = 16$ cm; $OM \perp AB$, $ON \perp CD$.
To find: which chord is nearer O; OM and ON.

1. $AB > CD$, so AB is nearer the centre (Theorem 8).
2. $AM = 15$ cm (Theorem 4). In right $\triangle OMA$, $OM^2 = 17^2 - 15^2 = 289 - 225 = 64$, so $OM = 8$ cm.
3. $CN = 8$ cm. In right $\triangle ONC$, $ON^2 = 17^2 - 8^2 = 289 - 64 = 225$, so $ON = 15$ cm.

Answer: AB is nearer the centre; it is 8 cm from O, and CD is 15 cm from O.

Example 3. Chords AB and CD of a circle with centre O are each 6 cm from O, and $AB = 16$ cm. Find CD and the radius of the circle.

FIG

Given: $OM \perp AB$, $ON \perp CD$, $OM = ON = 6$ cm, $AB = 16$ cm.
To find: CD and the radius OA.

1. The chords are equidistant from the centre, so $CD = AB = 16$ cm (Theorem 7).
2. $AM = 8$ cm (Theorem 4). In right $\triangle OMA$, $OA^2 = 6^2 + 8^2 = 100$, so $OA = 10$ cm.

Answer: $CD = 16$ cm and the radius is 10 cm.

Draw a rough figure for every question in Exercise 2B and mark all the given measurements on it.

Exercise 2B

1. A circle has centre O and radius 10 cm. Chords AB and CD are each 12 cm long. Find the distance of AB from O. Which theorem gives the distance of CD without a second calculation?
2. Chords PQ and RS of a circle with centre O are each 5 cm from O, and $PQ = 24$ cm. Find RS and the radius.
3. Use Pythagoras theorem to explain why Theorem 6 is true. Then use it to explain why Theorem 7 is true.
4. A circle has radius 13 cm. It has chords of lengths 10 cm, 24 cm and 26 cm. Without calculating, arrange them from the nearest to the farthest from the centre. Then find each distance to check.
5. In a circle of radius 5 cm, a chord of length 8 cm is 3 cm from the centre. Priya says, "A chord half as long, 4 cm, must be twice as far from the centre, 6 cm." Find her mistake.
6. A chord of a circle of radius r is at distance d from the centre. Explain why the length of the chord is $2\sqrt{r^2 - d^2}$.
7. A circular park has radius 17 m and a fountain at its centre. Two straight footpaths cross the park, each 30 m long from edge to edge.
   (a) How far is each footpath from the fountain?
   (b) The two paths are parallel and on opposite sides of the fountain. Find the distance between them.
8. Draw a circle in which a chord of length 6 cm is at a distance of 3 cm from the centre. Write your steps, and measure the radius.
9. True or false? Give a reason for each.
   (a) Two chords at the same distance from the centres of two different circles are equal.
   (b) Of two chords of a circle, the longer one is nearer the centre.
   (c) The chord farthest from the centre is the diameter.
10. In a circle with centre O, chords AB and AC are equal. Explain why the centre O lies on the bisector of $\angle BAC$.

5 Further questions

F1. Find the length of a chord of a circle of radius 7 cm that is 6 cm from the centre. Leave the answer in square-root form.

F2. Two equal chords AB and CD of a circle intersect at P. Show that $AP = CP$ and $PB = PD$.

F3. P is a point inside a circle with centre O (P is not O). Show that, of all the chords through P, the shortest one is perpendicular to OP.